

On virtual representations of finite Chevalley groups in defining characteristic

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G finite group, $\text{Rep}_k(G)$
 k -field, $k = \bar{k}$

1941, Brauer - Nesbitt defined the reduction map

$$K(\text{Rep}_{\mathbb{C}}(G)) \xrightarrow{\text{red}} K(\text{Rep}_k(G))$$

$\downarrow$
 k_p - char $p > 0$

$\begin{matrix} \rho \\ \downarrow \\ \text{over } \mathbb{C} \end{matrix}$ $\begin{matrix} \rho' \\ \downarrow \\ \mathbb{Z}_p \text{ (or } \mathcal{O}_K \text{ - finite over } \mathbb{Z}_p) \end{matrix}$

If $\rho = \rho' \otimes \mathbb{C}$
 $\mathcal{O}_K \subset \mathbb{C}$

$\rho \rightarrow \overline{\rho} = \rho' \otimes_{\mathcal{O}_K} k_p \quad (\rho' \bmod p)$

$K(\text{Rep}_{\mathbb{C}}(G)) \xrightarrow{\text{character}} \text{Cl}(G) - \text{class function on } G$

$\text{red} \downarrow$
 $K(\text{Rep}_{k_p}(G)) \xrightarrow{\text{Brauer character}} \text{Cl}(G_{(p)})$

$G_{(p)} = \{g \in G : (\text{ord}(g), p) = 1\}$

$\xrightarrow{?}$

$G = \underline{G}(\mathbb{F}_q)$, $\underline{G}$ - semisimple simply connected, split over $\mathbb{F}_q$

$K(\text{Rep}(\underline{G}_{\mathbb{C}})) \xrightarrow{\text{is}} K(\text{Rep}(\underline{G}_{\mathbb{F}_p})) \xrightarrow{\text{Weyl mod}} K(\text{Rep}_{k_p}(G))$ - is understood using character sheaves, etc.

$K(\text{Rep}_{k_p}(G))$

$\underline{G}(\mathbb{F}_q) \subset \underline{G}$

as an alg gp

$\underline{G}(\mathbb{F}_q)$

$\text{Rep}(\underline{G}) \xrightarrow{\text{res}} \text{Rep}_{k_p}(\underline{G}(\mathbb{F}_q))$

$K(\text{Rep}(\underline{G}_{\mathbb{C}})) \approx K(\text{Rep}(\underline{G}_{k_p}))$

$\downarrow \quad \downarrow$
 $K(\text{Rep}_{k_p}(G))$

Ex. from geometry

$$\begin{array}{ccc} X & \supset & G \\ \text{Smooth} & & \text{projective} \end{array}$$

$$P_1 = \mathbb{C}[X(\mathbb{F}_q)]$$

$\int_{\text{red}}$

$$\bar{P} = k_P[X(\mathbb{F}_q)] \xleftarrow{?}$$

$$\bar{P} \stackrel{?}{=} \tilde{P}|_{G(\mathbb{F}_q)}$$

$$\tilde{P} \in K(G_{k_P})$$

Answer: assuming splitting of diagonal for X

$$X(\mathbb{F}_q) = \Delta_X \cap \Gamma_{F_q}$$

graph of Frobenius

$$k_P[X(\mathbb{F}_q)] = \Gamma \left(\underset{\text{coh}(X \times X)}{\overset{\uparrow}{\mathcal{O}_\Delta}} \otimes \underset{\uparrow}{\mathcal{O}_{\Gamma_{F_q}}} \right)$$

$$\text{Suppose } [\mathcal{O}_\Delta] = \sum [\xi_i \boxtimes \xi_i^*]$$

$$\xi_i, \xi_i^* \in \text{coh}^4(X)$$

$$\text{Then get } \tilde{P} = \sum_i R\Gamma(\xi_i \otimes F^*(\xi_i^*))$$

$$\text{b/c in } K(\text{coh}^4(\mathbb{F}_q)(X^2)) : [\mathcal{O}_{\Gamma(F_q)}] = \sum [\xi_i \boxtimes F^*(\xi_i^*)]$$

$$\text{Ex. } X = \mathbb{P}^1, G = \text{SL}_2,$$

$$\mathcal{O}_{\mathbb{P}^1 \times \mathbb{P}^1}(-1, -1) \rightarrow \mathcal{O}_{\mathbb{P}^1 \times \mathbb{P}^1} \rightarrow \mathcal{O}_\Delta$$

$$F^*(\mathcal{O}(-1)) = \mathcal{O}(-p)$$

$$R\Gamma(\mathcal{O}) = k_P - \text{trivial}$$

$$[R\Gamma(\mathcal{O}(-p-1))] = -[R\Gamma(\mathcal{O}(p-1))]$$

$$\text{Answer: } [w_0] + [w_{p-1}]$$

Answer for p -unipotent rep of $G(\mathbb{F}_q)$ over $\mathbb{C}$.

Conj. (Lusztig 2001) (Here specialize to $G = SL(n)$)

unipotent rep'n = rep's gen'd by $B(\mathbb{F}_q)$ -inv. vectors $\longleftrightarrow$ partitions

$$W = \mathcal{O}_n = \bigsqcup_i \mathcal{C}_i$$

2-sided cells $\longleftrightarrow$ partitions

$$\mathcal{O}_n \xrightarrow{\text{RSK}} \text{partitions}$$

Fix partition λ , $\mathcal{S}_\lambda$

$$\begin{array}{ccc} \mathcal{S}_\lambda \cap \text{Inv} & \longleftrightarrow & \text{standard tableaux of shape } \lambda \longleftrightarrow \text{basis in irrep} \\ \parallel & & \\ \{w \in \mathcal{O}_n : w^2 = 1\} & & (\# = \dim \text{irrep}) \uparrow \mathcal{O}_n \end{array}$$

$w \in W = \mathcal{O}_n$, $L(w)$ - (left) descent set

$$\parallel$$

$$\{i : \ell(siw) < \ell(w)\}$$

$$\omega_w = \sum_{i \in L(w)} \omega_i$$

fund. weights

Conj. (Lusztig)

$\parallel$

Th (RB, Finkelberg, Kazhdan, Morton-Ferguson)

$$\exists \tilde{e}_\lambda = \sum_{w \in \mathcal{S}_\lambda \cap \text{Inv}} M_w$$

$$\kappa(\text{Rep}(G_{k,p})) = \kappa(\text{Rep}(G_{\mathbb{C}}))$$

M_w - sum of a few terms of the form $[w_{\mu_i}]$

μ_i - weight (close to $(p-1)w_w$).

Jantzen formula ($p > n$)

Rem p -regular elts in $G(\mathbb{F}_q)$ are ^{the} semisimpl elts $\uparrow \cong (q)$

$$\{p \text{ reg conj. classes}\} = T/W(\mathbb{F}_q) \simeq T/W(\mathbb{C})^{[q]}$$

$$\mathbb{F}_q^x \hookrightarrow \mathbb{C}^x$$

$$[q] + \text{mod } W \mapsto t^{q \text{ mod } W}$$

Ex. SL_2 , either $t = t^q$ or $t^{-1} = t^q$.

In this description can see the almost Brauer characters.

1) Character sheaves not of principal series

$\mathcal{F}$ have the property: $\mathcal{F}_g = 0$, g -semisimple

So ($p \gg 0$) Brauer reduction of the function vanishes

2) A principal series CS is a direct summand of $\widehat{\underline{G}} \subset \underline{G} \times \mathcal{B}^{\underline{G}/\underline{B}}$

$$W \sim \pi_X \underline{G} \quad \begin{array}{c} \pi \downarrow \\ \underline{G} \end{array}$$

$\psi \in \text{Irr}(W)$,

$$\mathcal{F}_\psi = [\pi_X \underline{G}, \psi]$$

$$(\underline{G} \approx \underline{G}_\epsilon)$$

For $g \in \underline{G}^{ss}$, $\pi^{-1}(g) = \coprod \mathcal{B}_i$

flag varieties of smaller gps

$$\text{Using this, } \pi_X \underline{G} \simeq (p_* \mathcal{O})_{\overline{g}}$$

$$F \xrightarrow{p} T/W$$

Conclusion

$$\psi \in \text{Irr}(\widehat{G}_n) \rightsquigarrow \mathcal{F}_\psi \xrightarrow{Tr(F_r)} \phi_\psi \in \text{CL}(G(\mathbb{F}_q))$$

(for $G = SL_n$, not in general, ϕ_ψ irr char. of $G(\mathbb{F}_q)$)

$$\omega(\phi_\psi) \in \text{Fun}(T/W(\mathbb{C})^{[1]})$$

$$\omega(\phi_\psi)(g) = \text{Tr}([1]^*, [(p_* \mathcal{O}_T)_g : \psi])$$

$$K(\text{Rep}(G_{\mathbb{C}})) \otimes \mathbb{C} \simeq \mathcal{O}(T/W) \xrightarrow[\text{pages}]{\omega} \mathcal{O}(T/W^{[1]})$$

Procedure: find $[\tilde{q}] : \mathcal{O}(T) \rightarrow \mathcal{O}(T)$ which

- 1 - $\mathcal{O}(T/W)$ - linear
- 2 - W -equiv. (on subquotient)
- 3 - agrees w/ $[q]^*$ on

then $\tilde{p}$ comes from $T_{\mathcal{O}(T)}([\tilde{q}]^*, [\mathcal{O}(T) : \psi])$ $P^{-1}((T/W)[q])$

Rem (Steinberg) $\mathcal{O}(T)$ is free over $\mathcal{O}(T/W)$

$[\mathcal{O}(T) : \psi]$ is free of rank $= \dim \psi = \# \{ \lambda \in \Lambda \cap I_{\text{inv}} \}$
 $\psi = \psi_{\lambda}$

M_{ψ} - diagonal entries of $[\tilde{q}]$ acting on $[\mathcal{O}(T) : \psi]$.

fix a basis t_x for $\mathcal{O}(T)$ over $\mathcal{O}(T/W)$

$[\tilde{q}]^*(t_x) = [q]^* t_x$ extend to $\mathcal{O}(T)$ by $\mathcal{O}(T/W)$ -lin.

$\S$ 1, 3 satisfied

- Steinberg described a basis

$\mathcal{O}(T)$ has $\mathbb{C}$ -basis e_{λ}

$e_{w(w_w)} -$ Steinberg basis

This doesn't satisfy (2), but (Kazhdan - Lusztig - Steinberg basis)

$\mathbb{Z}[W] \ni \mathbb{C}_w^{\pm}$, $t_w = \mathbb{C}_w^{\pm}(e_{w(w_w)}) -$ works

$\underbrace{\hspace{1cm}}$

Connection to geometry / decomposition of diagonal

This gives a lift $\tilde{p}$ for any unipotent rep'n ρ of $G(\mathbb{F}_q)$

eg. can take $\rho = \mathbb{C}[G/\rho(\mathbb{F}_q)]$

Th. This lift agrees w/ the one coming from decomposition of diagonal from KC Springer result.

$\mathcal{B} = \mathcal{G}/\mathcal{B} / k$ of char $p \gg 0$.

$\mathcal{D}_{\mathcal{B}}$ - Azumaya algebra on $T^* \mathcal{B}^{(1)}$, splits on the formal nbhd of $\mathcal{B}$
 $\mathcal{E}$ -splitting bundle, has $|W|$ -indecomposable summands $\mathcal{E}_{\omega}$

Also have dual object $\mathcal{E}^W \in D^b \text{Coh}(T^* \mathcal{B}^{(1)})$

$$\text{Ext}^{>0}(\mathcal{E}, \mathcal{E}) = 0, \quad A = \text{End}(\mathcal{E})^{\circ p}$$

$$D^b(\text{Coh } T^* \mathcal{B}) \simeq D^b(A\text{-mod})$$

$$\mathcal{E}|_{\mathcal{B}} \simeq F_* \mathcal{O}_{\mathcal{B}}$$

$$\text{Ext}^*(\mathcal{E}_{\omega}, \mathcal{E}^{\omega}) = k \delta^{\omega}$$

$\mathcal{E}_{\omega}, \mathcal{E}^{\omega}$ have G -equiv. structure

$$p: T^* \mathcal{B} \rightarrow \mathcal{B}$$

$$[\mathcal{O}_{\Delta}] = \sum [p_* \mathcal{E}^{\omega}] \boxtimes \mathcal{E}_{\omega}|_{\mathcal{B}} \quad (\text{and similar for } \mathcal{G}/\mathcal{P})$$

$$M_{\omega} = \left[R\Gamma \left(F_* (\mathcal{E}_{\omega}|_{\mathcal{B}}) \otimes p_* (\mathcal{E}^{\omega}) \right) \right]$$

$$\underline{L} \quad \Psi = \text{sign}, \quad [\mathcal{O}(T): \Psi] = \mathcal{O}(T/\omega) \delta$$

Weyl denominator

$$[\tilde{q}]^* [\delta] = [q]^* \delta$$

$$\tilde{\rho} = \frac{[q]^* \delta}{\delta} = \frac{\text{ch } V_{(q-1)\rho}}{\delta}$$

(B. Brena, Alvarez, McBreen, Yun) First kernel

$$K(\mathcal{G}_k\text{-mod}) \longrightarrow K(\mathcal{G}_1.T\text{-mod}) \xrightarrow{\sim} H_{\text{top}}(\text{Fl}_e)$$

affine Springer fibre for G^V

$$e = ts, \quad s \in \check{G}^V s$$

$$\in \mathcal{G}((t))$$

Labels

$G_2 T\text{-mod}$

baby Verma $\mathcal{M}_w$

$[M_w] \mapsto \text{limit cycle}$

Component $\rightsquigarrow$ image of
 $\text{gve} \rightsquigarrow \text{Fl}_e^{[M_w]}$
 $\downarrow$
 affine grassmannian